

Workflow and architecture patterns for working together 1: Workflows

Jacob Archambault

May 15, 2026

Outline

1 Current challenges and their canonical solutions

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - Benefits

Outline

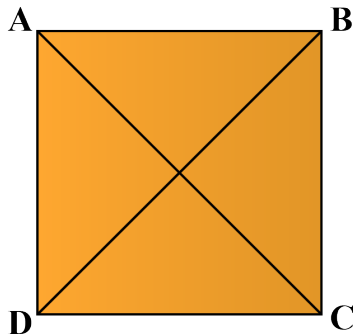
- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - Benefits
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - Benefits
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog
- 5 Conclusion

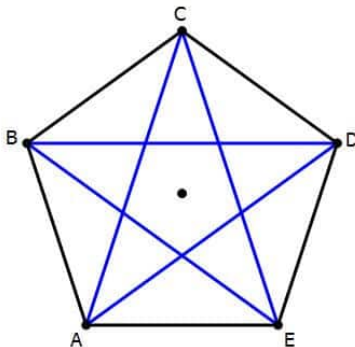
1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals= $n(n-1)/2$



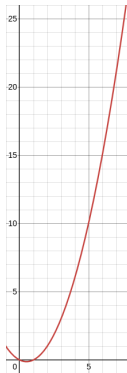
1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals= $\frac{n(n-1)}{2}$



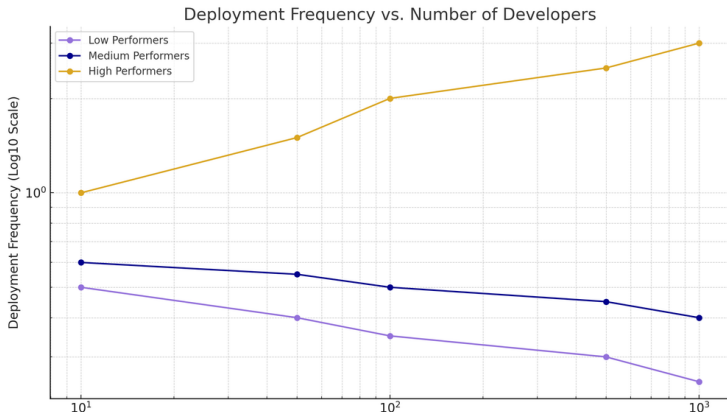
1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals=
$$\frac{n(n-1)}{2}$$



The Mythical Man Month, Fred Brooks (cont.)

- corollary: adding more people to a project can lead not only to diminishing returns on delivery speed, but to objectively less work being completed



Challenges

- coordinating multiple teams working in the same repository

Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing

Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code

Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code
 - feedback delays

Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code
 - feedback delays
 - merge conflict regressions

Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code
 - feedback delays
 - merge conflict regressions
 - sunk costs fallacy risks

Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code
 - feedback delays
 - merge conflict regressions
 - sunk costs fallacy risks
- coupling of all services at the database layer

Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code
 - feedback delays
 - merge conflict regressions
 - sunk costs fallacy risks
- coupling of all services at the database layer
 - high storage costs

Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code
 - feedback delays
 - merge conflict regressions
 - sunk costs fallacy risks
- coupling of all services at the database layer
 - high storage costs
 - single point of failure risk

Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code
 - feedback delays
 - merge conflict regressions
 - sunk costs fallacy risks
- coupling of all services at the database layer
 - high storage costs
 - single point of failure risk
 - risk of cross-project interference

Challenges

- coordinating multiple teams working in the same repository
- lack of direct collaboration, and occasionally trust, between development, operations, and testing
- late integration of contributor code
 - feedback delays
 - merge conflict regressions
 - sunk costs fallacy risks
- coupling of all services at the database layer
 - high storage costs
 - single point of failure risk
 - risk of cross-project interference

Canonical solutions to these challenges

- multiple teams → inverse Conway maneuver

Canonical solutions to these challenges

- multiple teams → inverse Conway maneuver
- collaboration → business stream-aligned teams over activity oriented teams

Canonical solutions to these challenges

- multiple teams → inverse Conway maneuver
- collaboration → business stream-aligned teams over activity oriented teams
- late integration → adopting a true continuous integration workflow

Canonical solutions to these challenges

- multiple teams → inverse Conway maneuver
- collaboration → business stream-aligned teams over activity oriented teams
- late integration → adopting a true continuous integration workflow
- database coupling → Bezos API mandate/“Microservices”

Canonical solutions to these challenges

- multiple teams → inverse Conway maneuver
- collaboration → business stream-aligned teams over activity oriented teams
- late integration → adopting a true continuous integration workflow
- database coupling → Bezos API mandate/“Microservices”

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - Benefits
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog
- 5 Conclusion

1967: Conway's law

'Organizations which design systems [...] are constrained to produce designs which are copies of the communication structures of these organizations.' - Melvin Conway, 'How do Committees Invent?' *Datamation*, 1967

Conway's law: examples

- 'If you have four groups working on a compiler, you'll get a 4-pass compiler' - The New Hacker's Dictionary, 1996

Conway's law: examples

- 'If you have four groups working on a compiler, you'll get a 4-pass compiler' - The New Hacker's Dictionary, 1996
- front-end [devs] business layer [backend devs] monolithic database [DBA team]

Conway's law: examples

- 'If you have four groups working on a compiler, you'll get a 4-pass compiler' - The New Hacker's Dictionary, 1996
- front-end [devs] business layer [backend devs] monolithic database [DBA team]
- a web api controller [manager] delegates most business logic to business classes [developers] which are part of the same in-memory process [team], while serving as a single entry-point for wider cross-network [cross-team] communication.

Conway's law: more examples

- separate .csproj projects for tests written by QA

Conway's law: more examples

- separate .csproj projects for tests written by QA
- separate development and config repositories for the same transaction signifying low communication between development and operations

Conway's law: more examples

- separate .csproj projects for tests written by QA
- separate development and config repositories for the same transaction signifying low communication between development and operations
- all business documentation in the same repository, separate from the repositories that the documentation is for

Conway's law: more examples

- separate .csproj projects for tests written by QA
- separate development and config repositories for the same transaction signifying low communication between development and operations
- all business documentation in the same repository, separate from the repositories that the documentation is for

Conway's law: corollaries

- Conway's law can't be fought

Conway's law: corollaries

- Conway's law can't be fought
- Teams working in the same part of the same codebase at the same time are the same team

Conway's law: corollaries

- Conway's law can't be fought
- Teams working in the same part of the same codebase at the same time are the same team
 - separate the codebases for improved velocity

Conway's law: corollaries

- Conway's law can't be fought
- Teams working in the same part of the same codebase at the same time are the same team
 - separate the codebases for improved velocity
 - join the teams for improved code sharing and communication (Inverse conway maneuver)

Conway's law: corollaries

- Conway's law can't be fought
- Teams working in the same part of the same codebase at the same time are the same team
 - separate the codebases for improved velocity
 - join the teams for improved code sharing and communication (Inverse conway maneuver)
 - branching-based strategies for team management compromise between the above approaches without their benefits

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - **Communication anti-pattern: local optimization**
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - Benefits
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog
- 5 Conclusion

communication anti-pattern: local optimization

- example: separate (DevOps, QA, operations, support) team

communication anti-pattern: local optimization

- example: separate (DevOps, QA, operations, support) team
- leads to bottlenecks

communication anti-pattern: local optimization

- example: separate (DevOps, QA, operations, support) team
- leads to bottlenecks
- delays communication

communication anti-pattern: local optimization

- example: separate (DevOps, QA, operations, support) team
- leads to bottlenecks
- delays communication
- discourages cross-functional collaboration

communication anti-pattern: local optimization

- example: separate (DevOps, QA, operations, support) team
- leads to bottlenecks
- delays communication
- discourages cross-functional collaboration
- punishes untracked work (e.g. thorough code reviews/dev testing, helping blocked colleagues)

Solution

- autonomous, cross-functional teams with 't-shaped' employees

Solution

- autonomous, cross-functional teams with 't-shaped' employees
- communities of practice

Solution

- autonomous, cross-functional teams with 't-shaped' employees
- communities of practice

Revisiting Conway's law: defining a team

- A *team* is the minimal socio-technical unit that can deliver a production change with no additional coordination beyond mandated governance or custodial controls.

Revisiting Conway's law: defining a team

- A *team* is the minimal socio-technical unit that can deliver a production change with no additional coordination beyond mandated governance or custodial controls.
 - pull requests

Revisiting Conway's law: defining a team

- A *team* is the minimal socio-technical unit that can deliver a production change with no additional coordination beyond mandated governance or custodial controls.
 - pull requests
 - testing

Revisiting Conway's law: defining a team

- A *team* is the minimal socio-technical unit that can deliver a production change with no additional coordination beyond mandated governance or custodial controls.
 - pull requests
 - testing
 - deployment

Revisiting Conway's law: defining a team

- A *team* is the minimal socio-technical unit that can deliver a production change with no additional coordination beyond mandated governance or custodial controls.
 - pull requests
 - testing
 - deployment

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - **Throughput**
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - Benefits
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog
- 5 Conclusion

Just-in-time manufacturing and Goldratt's theory of constraints

- Increase: throughput

Just-in-time manufacturing and Goldratt's theory of constraints

- Increase: throughput
- Decrease:

Just-in-time manufacturing and Goldratt's theory of constraints

- Increase: throughput
- Decrease:
 - operating costs

Just-in-time manufacturing and Goldratt's theory of constraints

- Increase: throughput
- Decrease:
 - operating costs
 - inventory

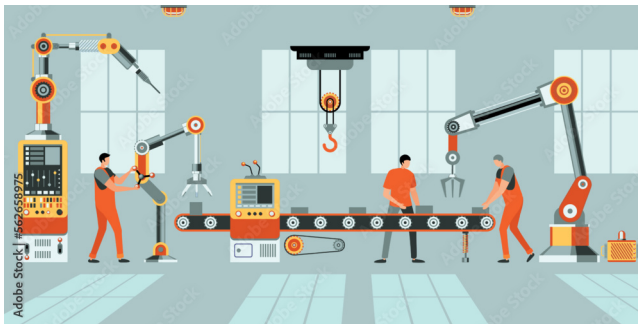
Just-in-time manufacturing and Goldratt's theory of constraints

- Increase: throughput
- Decrease:
 - operating costs
 - inventory
 - scrap

Just-in-time manufacturing and Goldratt's theory of constraints

- Increase: throughput
- Decrease:
 - operating costs
 - inventory
 - scrap
- Remove bottlenecks

Goldratt's theory of constraints (cont.)



2013: The State of DevOps Report

- a series of surveys of over 36,000 professionals

2013: The State of DevOps Report

- a series of surveys of over 36,000 professionals
- Led by a PhD statistician, Nicole Forsgren, from 2013-2019

2013: The State of DevOps Report

- a series of surveys of over 36,000 professionals
- Led by a PhD statistician, Nicole Forsgren, from 2013-2019
- Uses *cluster analysis* to discover groupings - has no prior understanding of what counts as good or bad.

The State of DevOps Report: throughput metrics

- lead time for changes

The State of DevOps Report: throughput metrics

- lead time for changes
- deployment frequency

The State of DevOps Report: stability metrics

- change failure rate

The State of DevOps Report: stability metrics

- change failure rate
- mean time to restore

The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.'
- Nicole Forsgren, Jez Humble, and Gene Kim, *Accelerate* (2017), p. 14

The State of DevOps Report: results

When compared to low performers, elite performers realize

127x

faster lead
time

182x

more
deployments
per year

8x

lower change
failure rate

2293x

faster failed
deployment
recovery times

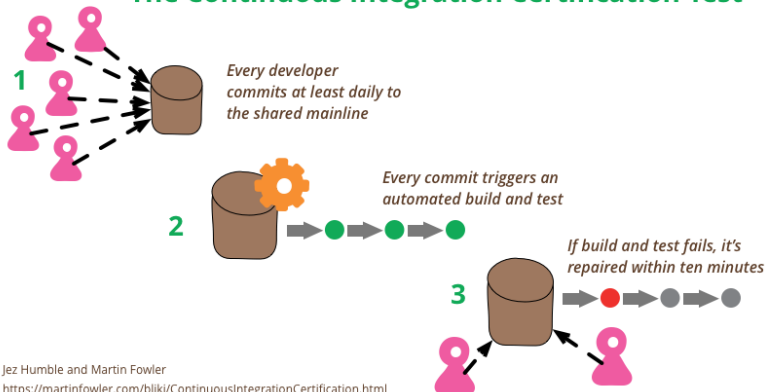
- 2024 State of DevOps Report

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - Benefits
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog
- 5 Conclusion

Defining continuous integration

The Continuous Integration Certification Test



Jez Humble and Martin Fowler

<https://martinfowler.com/bliki/ContinuousIntegrationCertification.html>

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - Benefits
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog
- 5 Conclusion

Continuous integration branching patterns

Continuous integration branching patterns

- don't

Continuous integration branching patterns

- don't
- as little as possible, for as short a time as possible, at least daily

Continuous integration branching patterns

- don't
- as little as possible, for as short a time as possible, at least daily

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - Benefits
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog
- 5 Conclusion

Continuous integration: trade-offs

Continuous Integration applies a different trigger for integration - you integrate whenever you've made a hunk of progress on the feature and your branch is still healthy. There's no expectation that the feature be complete, just that there's been a worthwhile amount of changes to the codebase. The rule of thumb is that "everyone commits to the mainline every day", or more precisely: you should never have more than a day's work sitting unintegrated in your local repository.

- Martin Fowler, "Branching Patterns"

Feedback comes late

Too late to change anything substantial

Low quality feedback

Through comments, lacking nuance

"My code"

Person owns area of code; key-person dependencies

Individual coding styles

Harder to have standards

Infrequent integration

Only integrate at the end of story/feature

Easy to ignore a failing build

Because it runs on an isolated branch

Dread large refactorings

Because they are likely to cause merge conflicts

People work in isolation

Harder to spot if someone needs help

Feedback comes in real time

As you're writing code or even before

Better quality feedback

Through face-to-face discussion

"Our code"

Collective code ownership

Team coding style

Easier to establish standards when pairing

Continuous Integration

Integrate as soon as code is pushed

We get used to not breaking things

Broken build on master is the team top priority

Easier to tackle large refactorings

As the rest of the team is always up to date

Visibility of what everyone is doing

Easier to spot if someone needs help

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - **Enablers**
 - Benefits
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog
- 5 Conclusion

Continuous integration: Enablers

- fast PR turnaround

Continuous integration: Enablers

- fast PR turnaround
- improving CI pipeline runtime

Continuous integration: Enablers

- fast PR turnaround
- improving CI pipeline runtime
- team discipline

Continuous integration: Enablers

- fast PR turnaround
- improving CI pipeline runtime
- team discipline

Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - **Benefits**
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog
- 5 Conclusion

Continuous integration: benefits

- High performers have the shortest integration times and branch lifetimes, with branch life and integration typically lasting hours.

Continuous integration: benefits

- High performers have the shortest integration times and branch lifetimes, with branch life and integration typically lasting hours.
- Low performers have the longest integration times and branch

Continuous integration: benefits

- High performers have the shortest integration times and branch lifetimes, with branch life and integration typically lasting hours.
- Low performers have the longest integration times and branch lifetimes, with branch life and integration typically lasting days.

Continuous integration: benefits

- High performers have the shortest integration times and branch lifetimes, with branch life and integration typically lasting hours.
- Low performers have the longest integration times and branch lifetimes, with branch life and integration typically lasting days.
- These differences are statistically significant
 - 2017 State of DevOps Report, p. 41

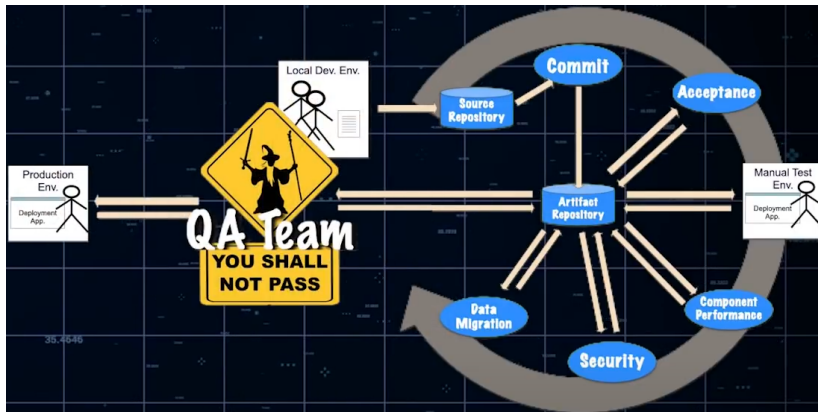
Outline

- 1 Current challenges and their canonical solutions
- 2 DevOps 201
 - Communication
 - Communication anti-pattern: local optimization
 - Throughput
- 3 Branching and continuous integration
 - Definition
 - CI Branching patterns
 - Trade-offs
 - Enablers
 - Benefits
- 4 Communication and throughput anti-patterns and solutions
 - Throughput anti-pattern: Gandalf vs. the Balrog
- 5 Conclusion

Throughput anti-pattern: Gandalf vs. the Balrog



Throughput anti-pattern: Gandalf vs. the Balrog



Throughput anti-pattern: Gandalf vs. the Balrog

- change approval boards

Throughput anti-pattern: Gandalf vs. the Balrog

- change approval boards
- pre-merge code reviews

Throughput anti-pattern: Gandalf vs. the Balrog

- change approval boards
- pre-merge code reviews
- 'bridge-to-nowhere' pipelines

Throughput anti-pattern: Gandalf vs. the Balrog

- change approval boards
- pre-merge code reviews
- 'bridge-to-nowhere' pipelines
- these all increase lead time

Throughput anti-pattern solutions: aggressively parallelize work

- pair programming

Throughput anti-pattern solutions: aggressively parallelize work

- pair programming
- parallelized pipeline builds for 'wide' pipelines

Throughput anti-pattern solutions: aggressively parallelize work

- pair programming
- parallelized pipeline builds for 'wide' pipelines
- one build artifact for all pipeline stages

Throughput anti-pattern solutions: aggressively parallelize work

- pair programming
- parallelized pipeline builds for 'wide' pipelines
- one build artifact for all pipeline stages
- share responsibility

Defining throughput for our project

- Avoid 'thrashing' metrics

Defining throughput for our project

- Avoid 'thrashing' metrics
- *Throughput* - the total lapsed time from local commit to production deployment

Defining throughput for our project

- Avoid 'thrashing' metrics
- *Throughput* - the total lapsed time from local commit to production deployment
- *Throughput** - the total lapsed time from local commit to user acceptance testing handoff

Defining throughput for our project

- Avoid 'thrashing' metrics
- *Throughput* - the total lapsed time from local commit to production deployment
- *Throughput** - the total lapsed time from local commit to user acceptance testing handoff

Conclusion

- Reduce goal misalignment

Conclusion

- Reduce goal misalignment
- stop people from waiting on each other

Conclusion

- Reduce goal misalignment
- stop people from waiting on each other
- work in small batches

Conclusion

- Reduce goal misalignment
- stop people from waiting on each other
- work in small batches
- That's mostly it

Conclusion

'DevOps is whatever you do to bridge friction created by silos, and all the rest is engineering' - Patrick Debois, Puppet State of DevOps report 2021

Workflow and architecture patterns for working together 1: Workflows

Jacob Archambault

May 15, 2026